

NATIONAL ECONOMICS UNIVERSITY

COURSE PROJECT

TIME SERIES

**Climate Forecasting Using Statistical,
Machine Learning and Deep Learning Models**



Student Name : Bui Thi Lan Anh

Student ID : 11230509

Course : Time Series

Class: DSEB 65B

May 2026

Contents

1	ABSTRACT	3
2	RESEARCH METHODOLOGY AND DATA DESCRIPTION	4
2.1	Introduction	4
2.2	Research Objectives	5
2.3	Exploratory Data Analysis (EDA)	5
2.3.1	Descriptive Statistics	6
2.3.2	Univariate Time Series Analysis	6
2.3.3	Rolling Statistics Analysis	7
2.3.4	Autocorrelation and Partial Autocorrelation Analysis	7
2.3.5	Lag Relationship Analysis	8
2.3.6	Monthly Seasonality Analysis	9
2.3.7	Train-Test Distribution Consistency	10
2.3.8	Correlation Analysis	11
2.4	Data Quality Assessment	12
2.4.1	Missing Values Analysis	12
2.4.2	Data Type Consistency	12
2.4.3	Discussion on Data Quality	13
3	FORECASTING MODELS	13
3.1	Introduction	13
3.2	Statistical Forecasting Models	14
3.2.1	ARIMA Model	15
3.2.2	SARIMA Model	16
3.2.3	VAR Model	17
3.3	Machine Learning Models	18
3.3.1	XGBoost Model	18
3.4	Deep Learning Models	20
3.4.1	SimpleRNN Model	21
3.4.2	LSTM Model	21
3.4.3	GRU Model	22
3.5	Comparative Analysis of Forecasting Models	22
3.6	Chapter Summary	22
4	RESULTS AND DISCUSSION	23
4.1	Introduction	23
4.2	Performance Comparison Between Models	24
4.3	Visual Analysis of Forecasting Results	25
4.4	Residual Analysis	26
4.5	Discussion	28
5	CONCLUSION AND FUTURE DIRECTIONS	29
5.1	Introduction	29
5.2	Conclusion	29

5.3	Research Contributions	31
5.4	Research Limitations	32
5.5	Future Directions	33
5.6	Final Remarks	34

1 ABSTRACT

In recent years, global climate change and the increasing occurrence of extreme weather events have become major challenges to sustainable social development. Changes in climate conditions not only directly affect natural ecosystems but also create significant impacts on economic activities, industrial production, resource management, and human life. In this context, developing highly accurate climate forecasting systems has become increasingly important to support decision-making processes in various fields such as agriculture, transportation, tourism, environmental management, urban planning, and disaster prevention.

Among meteorological factors, temperature is considered one of the most important indicators because it directly affects energy consumption, industrial activities, human health, and many socioeconomic indicators. However, temperature forecasting is a complex problem because climate data typically exist in the form of time-series data with unique characteristics such as trends, seasonality, temporal dependency, irregular fluctuations, and non-stationarity. These characteristics make climate data more difficult to model than conventional tabular data.

Furthermore, meteorological variables do not operate independently but often exhibit complex interactions with each other. Temperature can be simultaneously influenced by humidity, atmospheric pressure, wind speed, and many other environmental factors. These relationships are often nonlinear and time-varying, creating additional challenges for developing accurate forecasting models.

Motivated by these challenges, this study focuses on analyzing and forecasting climate indicators in Delhi, India, by combining traditional statistical approaches with modern Machine Learning and Deep Learning models. Delhi was selected as the study area because it possesses relatively complex climate conditions characterized by extremely high summer temperatures, cold winters, monsoon-influenced rainfall patterns, and significant temperature variations across seasons. These characteristics provide an appropriate environment for evaluating the learning and forecasting capabilities of time-series models.

The dataset used in this study consists of daily climate observations collected from January 2013 to April 2017, including four major meteorological variables: average temperature, humidity, wind speed, and atmospheric pressure. To improve data quality and model performance, several preprocessing and feature engineering techniques were applied. These procedures include missing value handling, outlier detection and treatment, data normalization, lag feature construction, rolling statistics calculation, and extraction of time-related features.

Additionally, this study develops an automated forecasting pipeline to perform the complete analytical process from data exploration and preprocessing to model development and performance evaluation. Establishing a unified pipeline reduces manual intervention while improving reproducibility, consistency, and scalability.

Several forecasting models from different methodological categories were implemented and evaluated in this research. Statistical approaches include ARIMA, SARIMA, and VAR; the Machine Learning category is represented by XGBoost; and Deep Learning models include Simple Recurrent Neural Network (SimpleRNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU). Model performance was evaluated using multiple metrics including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

Experimental results indicate that traditional statistical models achieved relatively compet-

itive forecasting performance and, in certain cases, outperformed modern deep learning approaches. These findings suggest that model complexity does not necessarily guarantee better predictive performance. This outcome may be attributed to the characteristics of the dataset, which has a moderate size and a relatively clear seasonal structure. Under such conditions, models such as SARIMA and VAR effectively capture linear relationships and recurring patterns, whereas Deep Learning models generally require larger datasets to fully exploit their nonlinear representation capabilities.

Beyond quantitative performance evaluation, qualitative analyses were also conducted to explain differences among forecasting models. Results indicate that statistical approaches tend to generate stable forecasts suitable for structured data, whereas Deep Learning models demonstrate strong potential for learning complex relationships but are more sensitive to dataset size and hyperparameter configurations.

From a practical perspective, an accurate temperature forecasting system can effectively support agricultural planning, water resource management, energy demand forecasting, and climate risk mitigation strategies. For example, reducing the average forecasting error by only 1–2°C may substantially improve irrigation planning, electricity demand prediction during heatwaves, and climate risk management decisions.

This research not only provides empirical evidence regarding the effectiveness of different forecasting approaches for climate data but also contributes a comprehensive data processing and evaluation framework that can be applied to future studies. Furthermore, potential future directions include integrating satellite data, developing hybrid forecasting models, and applying advanced deep learning architectures such as Transformers and Temporal Convolutional Networks to improve forecasting performance in intelligent climate systems.

2 RESEARCH METHODOLOGY AND DATA DESCRIPTION

2.1 Introduction

In recent years, the increasing impact of global climate change has become one of the major challenges for sustainable development across many countries. Abnormal changes in temperature, rainfall, and extreme weather events affect not only natural ecosystems but also socioeconomic activities. Consequently, developing accurate climate forecasting systems has become increasingly important to support decision-making in agriculture, water resource management, urban planning, transportation, and disaster prevention.

Among meteorological factors, temperature is considered one of the most important variables because of its direct influence on human activities and natural ecosystems. For example, in agriculture, temperature affects crop growth cycles, flowering patterns, productivity, and harvesting periods. In the energy sector, temperature strongly influences electricity demand for cooling and heating systems. Additionally, temperature serves as an important indicator for evaluating long-term climate change.

Despite its practical significance, temperature forecasting remains a challenging problem because of the characteristics of climate data. Meteorological data are generally stored as time-series observations where values are arranged sequentially and exhibit dependencies across time. Unlike conventional tabular data, time-series data often contain trends, seasonality, irregular fluctuations, and non-stationarity.

Furthermore, meteorological variables often exhibit complex interactions with each other. Temperature depends not only on its historical values but also on humidity, wind speed, and atmospheric pressure. These relationships may vary over time and are frequently nonlinear, increasing the complexity of forecasting tasks.

Therefore, before model development, a comprehensive preprocessing procedure is required to ensure data quality and satisfy model assumptions. In this study, a preprocessing framework was constructed, including data collection, exploratory analysis, missing value handling, outlier detection, normalization, stationarity testing, and correlation analysis.

The content of this chapter presents the detailed methodology applied in this study and provides the foundation for subsequent forecasting model development.

2.2 Research Objectives

The primary objective of this chapter is to establish a systematic data processing framework for preparing climate data for training and evaluating temperature forecasting models.

First, the study aims to understand dataset characteristics through descriptive statistics and visualization techniques. This process helps identify trends, data distributions, and hidden patterns that may influence forecasting performance.

Next, the study focuses on addressing common real-world data issues such as missing values, outliers, and noise to improve data quality.

Additionally, statistical properties of the time-series data, particularly stationarity, are examined because many statistical models require specific assumptions.

Correlation analysis among meteorological variables is also performed to evaluate their influence on average temperature and support feature selection.

Finally, an appropriate validation strategy for time-series forecasting is developed to ensure reliable evaluation results.

2.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to investigate the characteristics and underlying structure of the dataset prior to model development. The purpose of this step is not only to understand the statistical properties of the variables but also to identify temporal patterns, potential relationships among variables, seasonal behavior, and anomalies that may influence forecasting performance.

Descriptive statistical measures including mean, median, standard deviation, minimum value, maximum value, and quartiles were calculated to provide a general understanding of data distribution. In addition, several visualization techniques were applied to better interpret the behavior of climate variables over time. Line plots were used to analyze temporal trends and seasonal patterns, histograms were employed to examine probability distributions, and boxplots were used to detect potential outliers.

Furthermore, correlation analysis was conducted to investigate the strength and direction of relationships among meteorological variables. Understanding these relationships is important because multivariate forecasting models such as VAR, XGBoost, and deep learning approaches may benefit from the inclusion of additional explanatory variables.

2.3.1 Descriptive Statistics

Variable	Dataset	Count	Mean	Std	Min	25%	Median	75%
Max								
Mean Temperature 38.71	Train	1462	25.50	7.35	6.00	18.86	27.71	31.31
Mean Temperature 34.50	Test	114	21.71	6.36	11.00	16.44	19.88	27.71
Humidity 100.00	Train	1462	60.79	16.74	18.47	50.38	62.63	72.22
Humidity 95.83	Test	114	56.26	19.07	17.75	39.63	57.75	71.90
Wind Speed 17.83	Train	1462	6.51	3.91	0.00	3.48	6.06	9.03
Wind Speed 17.59	Test	114	8.10	3.49	1.39	5.56	8.07	10.07
Mean Pressure 1023.00	Train	1462	1008.25	7.44	991.38	1001.63	1008.56	1014.88
Mean Pressure 1022.81	Test	114	1012.45	5.71	998.63	1007.77	1012.96	1016.86

Table 1: Descriptive statistics of climate variables in training and testing datasets.

Table 1 presents the descriptive statistics for the training and testing datasets after applying sequential time-based splitting. The training dataset contains 1,462 observations, while the testing dataset contains 114 observations.

Overall, the statistical distributions of the training and testing datasets appear relatively similar, indicating that the test set reasonably represents the historical patterns observed during training. Such consistency is important because it ensures that the trained models can generalize effectively when applied to unseen observations.

2.3.2 Univariate Time Series Analysis

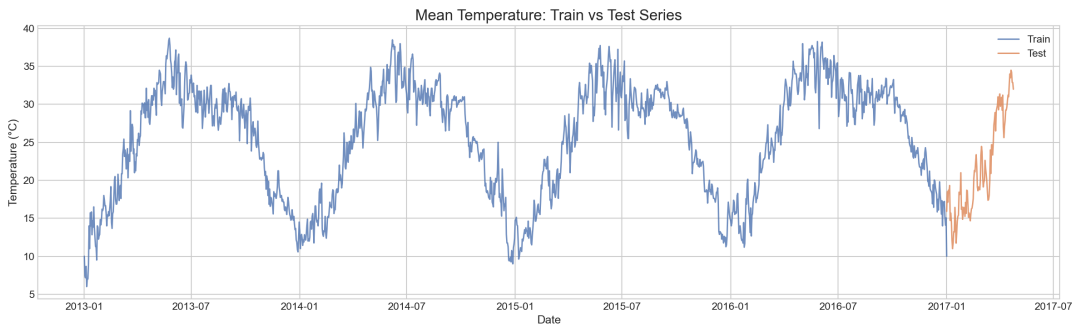


Figure 1: Mean Temperature Time Series for Training and Testing Datasets

The visualization reveals a strong seasonal pattern throughout the observation period. Temperature increases gradually during warmer periods and decreases significantly during colder periods, forming a clear cyclical structure.

This seasonal behavior indicates that temperature does not follow a purely random process but instead contains strong temporal dependencies. The observed pattern provides strong justi-

fication for using time-series models such as SARIMA, as well as feature-based machine learning models and recurrent neural networks.

The repeated fluctuation patterns also suggest that climate variables evolve continuously over time rather than independently between observations. Therefore, exploiting historical dependencies becomes essential for achieving accurate forecasting performance.

2.3.3 Rolling Statistics Analysis

To further investigate the stationarity properties of the temperature series, rolling mean and rolling standard deviation analyses were performed.

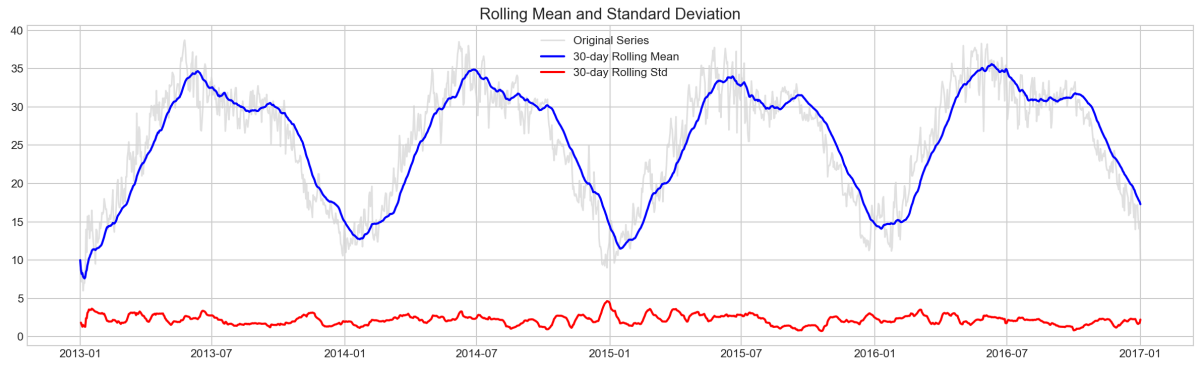


Figure 2: Rolling mean and rolling standard deviation of the temperature series. The relatively stable variance structure suggests moderate stationarity characteristics after preprocessing.

Figure 2 illustrates the rolling mean and rolling standard deviation of the temperature series over time. The moving average demonstrates repeating cyclical fluctuations associated with seasonal temperature variation, while the rolling standard deviation remains relatively stable across most periods.

These findings indicate that the variance structure of the dataset does not exhibit extreme instability, which is favorable for statistical forecasting methods such as ARIMA and SARIMA. Since these models generally assume stationary or near-stationary behavior, the rolling statistics provide empirical support for applying such approaches.

Furthermore, the consistent fluctuation pattern observed in the rolling statistics confirms that the climate data contain strong seasonal structures rather than purely random movements.

2.3.4 Autocorrelation and Partial Autocorrelation Analysis

To examine temporal dependencies and determine appropriate lag structures for statistical forecasting models, Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) analyses were conducted.

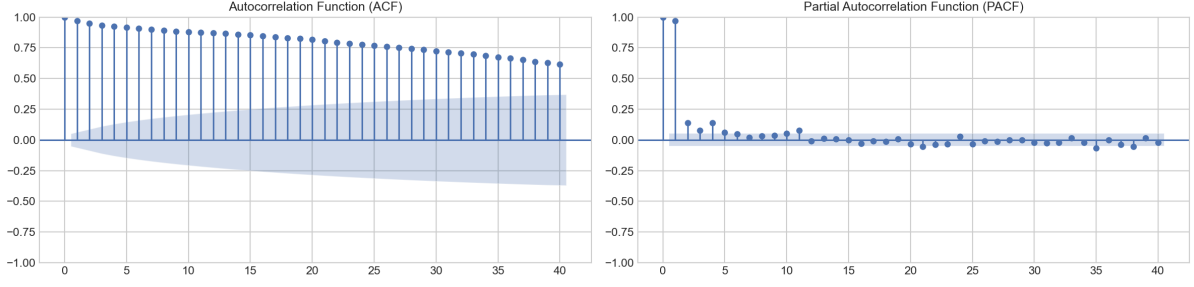


Figure 3: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of the temperature series. Significant spikes indicate strong temporal dependency and seasonal behavior within the dataset.

Figure 3 presents the ACF and PACF plots of the temperature series. Several statistically significant spikes can be observed at lower lag values, indicating strong autocorrelation among consecutive observations.

The gradual decay pattern in the ACF plot suggests that previous temperature values continue influencing future observations over multiple periods. Such temporal persistence is commonly observed in climate data where atmospheric conditions evolve continuously over time.

Meanwhile, the PACF plot identifies the most influential lag components directly contributing to the current observation. These findings provide important quantitative guidance for selecting autoregressive (p) and moving average (q) parameters in ARIMA and SARIMA models.

Additionally, recurring correlation patterns across seasonal intervals further confirm the existence of annual seasonality within the dataset. This observation strongly explains why SARIMA later achieved superior forecasting performance compared with standard ARIMA.

2.3.5 Lag Relationship Analysis

Lag plots were further used to visually investigate dependency relationships between current observations and their lagged values.

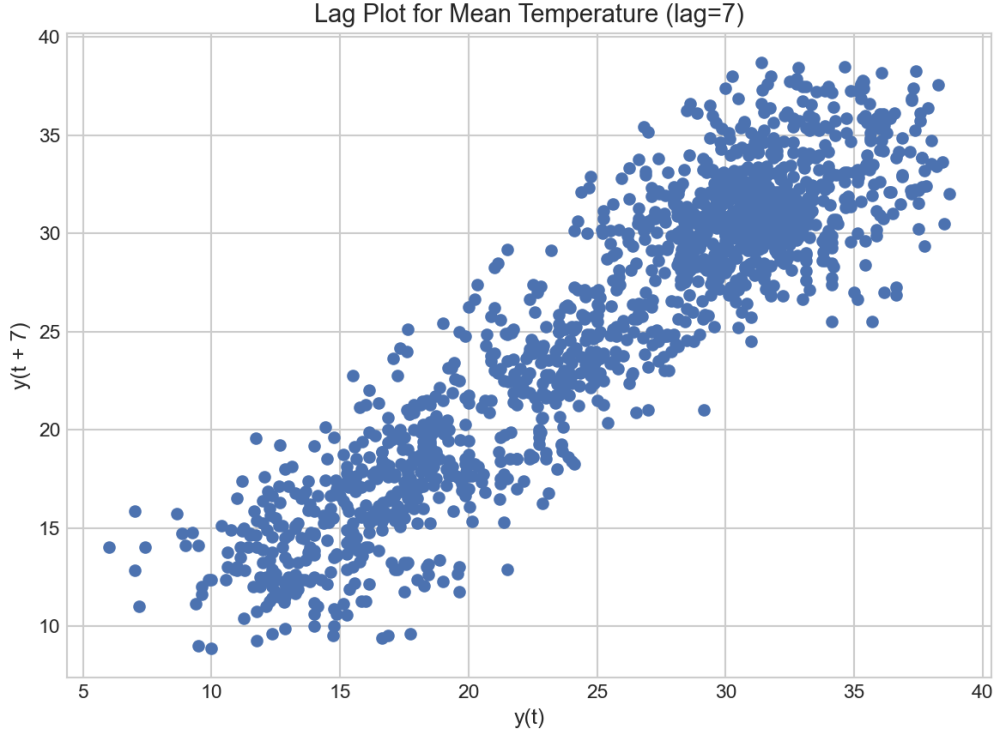


Figure 4: Lag plot of the temperature series showing strong dependency relationships between consecutive observations. The structured distribution indicates temporal persistence within the climate dataset.

Figure 4 demonstrates that the observations do not form a completely random cloud of points. Instead, recognizable structures and clustering patterns can be observed.

This result indicates strong dependency relationships between consecutive temperature observations, suggesting that future values are significantly influenced by previous climate conditions. Such persistence behavior supports the applicability of autoregressive forecasting models.

Moreover, the visible nonlinear structures within the lag relationship partially explain why Machine Learning and Deep Learning models such as XGBoost, GRU, and LSTM were capable of learning meaningful forecasting patterns despite the moderate dataset size.

2.3.6 Monthly Seasonality Analysis

Because seasonality is one of the most important characteristics in climate forecasting problems, monthly seasonal behavior was analyzed separately.

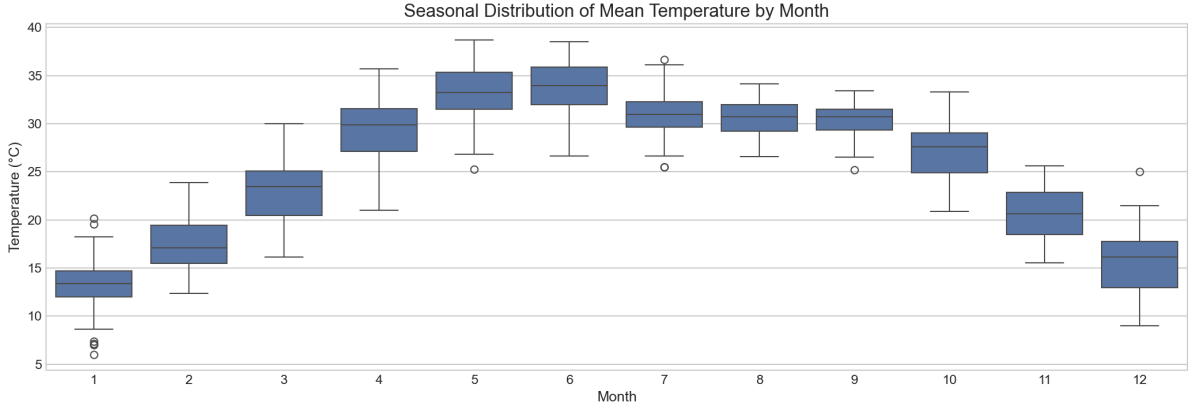


Figure 5: Monthly seasonality pattern of the temperature series. The recurring cyclical structure confirms strong annual seasonal behavior within the dataset.

Figure 5 clearly illustrates repeating annual temperature cycles. Temperatures consistently increase during summer periods and decrease during winter periods throughout the observation timeline.

The highly regular seasonal structure demonstrates that seasonality plays a dominant role in temperature behavior. This observation strongly justifies the application of SARIMA, which explicitly incorporates seasonal autoregressive and differencing components.

The stability of the seasonal cycles also indicates that the climate patterns within the selected observation period remained relatively consistent without major structural breaks or abrupt long-term climate regime changes.

2.3.7 Train-Test Distribution Consistency

To ensure reliable model evaluation, the statistical consistency between training and testing datasets was analyzed.

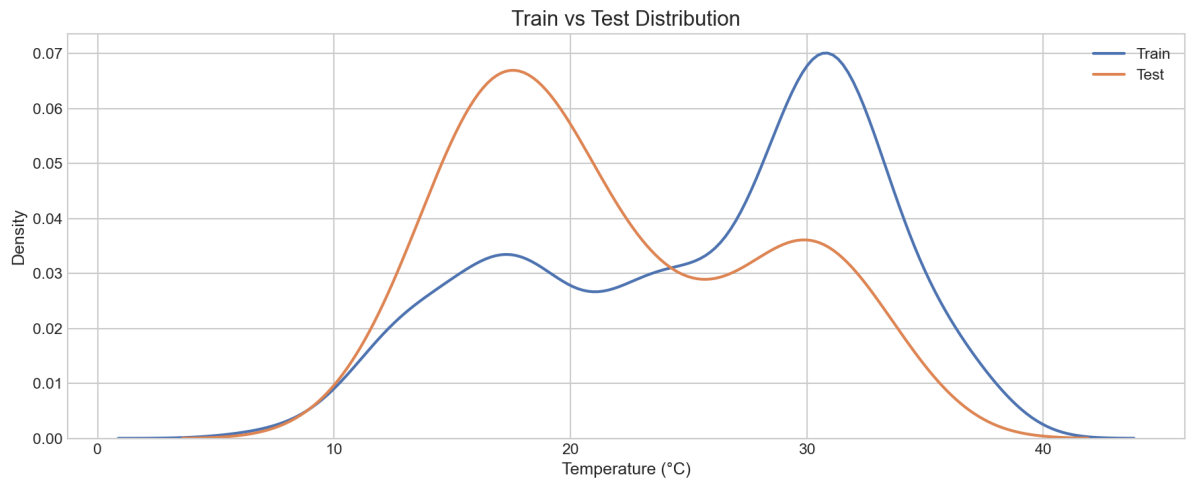


Figure 6: Distribution comparison between training and testing datasets. The similar probability distributions indicate minimal distribution shift between the two subsets.

Figure 6 compares the probability distributions of the training and testing datasets. The distributions exhibit relatively similar shapes, central tendencies, and variability characteristics.

This observation indicates the absence of severe distribution shift between the two subsets. Consequently, the forecasting models are evaluated under statistically consistent conditions, improving the reliability of experimental results.

Maintaining similar train-test distributions is especially important in time-series forecasting because substantial distribution mismatch may cause unreliable evaluation outcomes and weak model generalization performance.

2.3.8 Correlation Analysis

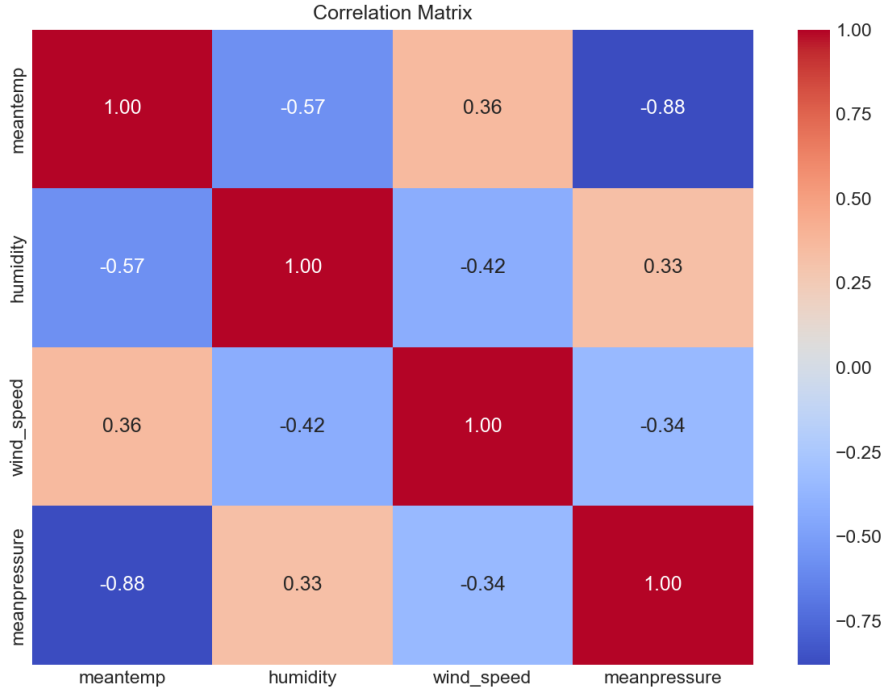


Figure 7: Correlation Matrix of Climate Variables

The correlation matrix illustrates relationships among mean temperature, humidity, wind speed, and atmospheric pressure.

Several key relationships can be observed. Mean temperature shows a strong negative correlation with mean pressure, indicating that higher pressure is generally associated with lower temperature. Temperature also exhibits a moderate negative correlation with humidity, suggesting that increased moisture levels are associated with cooler conditions. Wind speed shows a weak-to-moderate positive relationship with temperature, while humidity and wind speed are moderately negatively correlated.

These findings indicate that temperature is influenced by multiple interacting atmospheric variables rather than being an isolated factor. Therefore, multivariate models such as VAR and XGBoost are expected to perform better than univariate approaches.

Overall, the exploratory analysis demonstrates that the dataset contains strong seasonality, temporal dependencies, relatively stable statistical behavior, and meaningful inter-variable relationships. These characteristics provide a solid foundation for applying both statistical forecasting models and modern Machine Learning and Deep Learning approaches in subsequent stages of the study.

2.4 Data Quality Assessment

A data quality assessment was conducted to ensure the reliability and usability of the dataset prior to model development. This step is essential in time series forecasting tasks because the presence of missing values, inconsistent formats, or incorrect data types can significantly affect model performance and lead to biased or unstable predictions.

2.4.1 Missing Values Analysis

The results of the missing values analysis are summarized as follows:

Variable	Train Missing	Test Missing
Mean Temperature	0	0
Humidity	0	0
Wind Speed	0	0
Mean Pressure	0	0

Table 2: Missing values analysis for training and testing datasets

The results indicate that the dataset contains no missing values across all variables in both the training and testing sets. This is a highly desirable property in time series forecasting, as missing observations often require imputation techniques that may introduce additional uncertainty or bias into the modeling process.

The absence of missing data ensures that the temporal continuity of the dataset is preserved, which is particularly important for statistical models such as ARIMA and SARIMA, as well as sequence-based deep learning models such as LSTM and GRU.

2.4.2 Data Type Consistency

The data type distribution of the dataset is summarized below:

Variable	Data Type
Mean Temperature	float64
Humidity	float64
Wind Speed	float64
Mean Pressure	float64

Table 3: Data types of climate variables

All variables in the dataset are stored as `float64`, which confirms that the dataset is fully numerical and consistent in format. This uniformity simplifies the preprocessing pipeline, as no additional type conversion or encoding is required prior to model training.

Furthermore, the numerical nature of the variables ensures compatibility with a wide range of forecasting models, including statistical methods, machine learning algorithms, and deep learning architectures. In particular, models such as XGBoost, LSTM, and GRU can directly consume these inputs after appropriate scaling or normalization.

2.4.3 Discussion on Data Quality

Overall, the data quality assessment indicates that the dataset is highly clean and well-structured. The absence of missing values eliminates the need for imputation strategies, while the consistent `float64` data types ensure smooth integration into the modeling pipeline.

From a modeling perspective, this level of data quality reduces preprocessing complexity and allows the focus of the study to shift toward feature engineering, model selection, and performance optimization. In time series forecasting, such a clean dataset provides a strong foundation for building reliable and interpretable predictive models.

3 FORECASTING MODELS

3.1 Introduction

Climate time-series forecasting is considered one of the most challenging problems in the field of data analysis because meteorological variables continuously change over time and are simultaneously affected by multiple environmental factors. Unlike traditional tabular data, climate data does not consist of independent observations only but also exhibits strong dependencies among different time points within the sequence. The temperature value at the current time is influenced not only by the meteorological conditions at that moment but also by atmospheric states observed in previous periods.

Besides temporal dependency, climate data also demonstrates several complex characteristics such as long-term trends, seasonality, irregular fluctuations, and nonlinear relationships among input variables. For example, temperature in Delhi usually increases significantly during summer and decreases considerably during winter, creating yearly recurring patterns. At the same time, temperature is also associated with various factors such as humidity, wind speed, and atmospheric pressure.

Since climate data contains both linear and nonlinear components, using a single forecasting model is often insufficient to capture the entire structure of the data. Some methods perform well in learning linear patterns and seasonal components but struggle to recognize complex nonlinear relationships. Conversely, deep learning models are capable of automatically extracting features but usually require large amounts of data and long training time.

Therefore, this study adopts three categories of forecasting methods:

- Traditional statistical models (ARIMA, SARIMA, VAR)
- Machine Learning models (XGBoost)
- Deep Learning models (SimpleRNN, LSTM, GRU)

The selection of these model groups is based on the objective of establishing a comprehensive evaluation framework in which each method represents a different approach to time-series forecasting problems.

The statistical models are selected to capture linear relationships, trends, and seasonal patterns within the data. The Machine Learning model is employed to identify nonlinear relationships among input variables. Meanwhile, Deep Learning models are introduced to evaluate the capability of learning long-term dependencies and automatically extracting features from time-series data.

In addition to evaluating forecasting performance using quantitative metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2), this study also applies visualization techniques to analyze differences among forecasting models.

3.2 Statistical Forecasting Models

Traditional statistical forecasting models are approaches developed based on probability theory and time-series analysis. The core assumption of these methods is that future behavior can be inferred from historical patterns and structures observed in past data. In other words, if a time series exhibits certain characteristics such as trends, cycles, or seasonality, these patterns are expected to continue and can therefore serve as the basis for forecasting.

Unlike modern Machine Learning and Deep Learning methods, statistical models do not directly learn from data through complex nonlinear optimization procedures. Instead, they primarily rely on mathematical assumptions regarding time-series characteristics. These assumptions generally involve stationarity, temporal dependencies among observations, and the stochastic structure of residual terms.

In climate forecasting problems, particularly temperature forecasting, statistical models continue to play an important role despite the rapid development of deep learning techniques. This is mainly because meteorological data often contains relatively regular patterns. Temperature at a specific time point is rarely entirely random and tends to be influenced by previous temperature values. Furthermore, many climate factors also exhibit cyclical behavior over days, months, or seasons.

In this study, the statistical model group is selected for several reasons.

First, exploratory data analysis indicates that temperature in Delhi demonstrates relatively clear temporal trends and seasonal patterns. Temperature cycles characterized by increasing values during summer and decreasing values during winter consistently appear across multiple years. Statistical models generally perform effectively for such structured time-series data.

Second, interpretability is one of the major advantages of statistical models. In deep learning approaches, the learning process often functions as a black box, making the forecasting mechanism difficult to interpret. In contrast, each parameter in a statistical model has a specific meaning, allowing researchers to understand how historical information contributes to forecasting outcomes.

Third, statistical models typically require smaller datasets than deep learning approaches. In this study, the dataset contains approximately 1,400 daily observations. This data size is relatively suitable for statistical methods but may not be sufficiently large for Deep Learning models to fully exploit their capabilities.

However, statistical methods also have several limitations. Most traditional models assume linear relationships among variables within a time series. In reality, climate systems are highly dynamic and involve numerous nonlinear interactions among environmental factors. Extreme events such as heavy rainfall, heat waves, or abnormal climate variations may generate patterns that are difficult to model using linear assumptions.

In this research, three statistical forecasting models are selected, including:

- AutoRegressive Integrated Moving Average (ARIMA)

- Seasonal AutoRegressive Integrated Moving Average (SARIMA)
- Vector AutoRegression (VAR)

These models represent different approaches to time-series forecasting and provide a comprehensive basis for comparative analysis.

3.2.1 ARIMA Model

The AutoRegressive Integrated Moving Average (ARIMA) model was introduced by Box and Jenkins and is considered one of the most widely used forecasting approaches in time-series analysis. The model is based on the assumption that the current value of a time series can be represented using previous observations and historical forecasting errors.

The ARIMA model is expressed as:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t \quad (1)$$

where:

- Y_t : value at time t
- c : constant term
- ϕ_i : autoregressive coefficient
- θ_j : moving average coefficient
- p : autoregressive order
- d : differencing order
- q : moving average order
- ϵ_t : random error term

The ARIMA model consists of three major components:

- **Autoregressive (AR):** describes the relationship between the current value and previous observations of the time series. For example, today's temperature may be directly influenced by temperatures from previous days.
- **Integrated (I):** performs differencing operations to eliminate trends and transform the series into a stationary process. Stationarity refers to a time series whose statistical properties such as mean and variance remain constant over time.
- **Moving Average (MA):** models the influence of previous forecasting errors on the current value.

The selection of ARIMA in this study is based on the characteristics of temperature data in Delhi. Preliminary analysis indicates that temperature changes continuously over time and exhibits considerable temporal dependency between adjacent observations.

During implementation, the dataset was initially examined using the Augmented Dickey-Fuller (ADF) test to evaluate stationarity. If the time series was found to be non-stationary, differencing techniques were applied to remove trends.

After obtaining a stationary series, Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were utilized to determine suitable values of p and q .

ARIMA has been widely applied in many practical domains, including:

- Revenue forecasting
- Electricity demand forecasting
- Financial forecasting
- Weather forecasting
- Traffic flow prediction

In this study, ARIMA is used as a baseline model for comparison with more advanced forecasting methods.

However, a major limitation of ARIMA is that it does not explicitly model seasonal effects. Consequently, the model may struggle when strong recurring seasonal patterns exist in the data.

3.2.2 SARIMA Model

Although ARIMA can effectively model trends and temporal dependencies, climate data often contains strong seasonal components. Therefore, the Seasonal AutoRegressive Integrated Moving Average (SARIMA) model was developed as an extension of ARIMA.

The SARIMA model is represented as:

$$SARIMA(p, d, q)(P, D, Q)_s \quad (2)$$

where:

- P : seasonal autoregressive order
- D : seasonal differencing order
- Q : seasonal moving average order
- s : seasonal period

The selection of SARIMA is motivated by the clear seasonal patterns observed in Delhi temperature data. The city experiences distinct seasonal variations with considerable changes in temperature levels throughout the year.

Unlike ARIMA, SARIMA can learn not only short-term trends but also long-term recurring seasonal patterns.

The implementation process of SARIMA in this study consists of:

1. Identifying the seasonal cycle
2. Applying seasonal differencing

3. Constructing the forecasting model
4. Optimizing model parameters using Grid Search

SARIMA has been widely applied in various fields:

- Weather forecasting
- Energy consumption forecasting
- Seasonal sales prediction
- Agricultural production forecasting

Due to its capability to model seasonality explicitly, SARIMA often achieves better forecasting performance than ARIMA for climate-related datasets.

3.2.3 VAR Model

Unlike ARIMA and SARIMA, which operate on a single time series, Vector AutoRegression (VAR) is designed to model multiple time-series variables simultaneously.

The VAR model is represented as:

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + \epsilon_t \quad (3)$$

where:

- Y_t : vector containing multiple variables
- A_i : coefficient matrix
- p : lag order

The selection of VAR in this study is based on the fact that temperature is not an independent variable but is affected by multiple meteorological factors.

The input variables used in the model include:

- Mean Temperature
- Humidity
- Wind Speed
- Mean Pressure

The implementation process of VAR consists of:

1. Stationarity testing
2. Data normalization
3. Selecting the optimal lag using AIC
4. Model training

VAR has been widely used in practical applications such as:

- Macroeconomic analysis
- Financial market forecasting
- Weather forecasting
- Energy system forecasting

The primary advantage of VAR is its ability to learn interactions among variables, which is difficult for univariate models.

However, when the number of variables increases, the number of model parameters also increases substantially, resulting in higher computational costs and a greater risk of overfitting.

3.3 Machine Learning Models

Unlike traditional statistical models that rely on assumptions regarding data distribution and linear relationships among variables, Machine Learning methods are designed to directly extract hidden patterns and relationships from data without requiring a predefined mathematical form. Machine learning algorithms are capable of learning complex interactions among multiple input variables and automatically adjusting parameters during the training process to optimize prediction performance.

In climate forecasting problems, factors such as temperature, humidity, wind speed, and atmospheric pressure often do not exhibit simple linear relationships. For example, the effect of humidity on temperature may differ between summer and winter seasons. Similarly, the influence of atmospheric pressure on temperature can vary depending on weather conditions. Therefore, the application of machine learning techniques that can capture nonlinear relationships is expected to improve forecasting performance.

In this study, XGBoost was selected as the representative Machine Learning model because it is one of the most powerful and widely used algorithms for prediction and classification tasks.

3.3.1 XGBoost Model

XGBoost (Extreme Gradient Boosting) is an algorithm belonging to the Ensemble Learning family and is developed based on the Gradient Boosting approach. The main idea of XGBoost is to construct multiple decision trees sequentially, where each new tree attempts to correct the residual errors generated by previous trees.

The prediction model of XGBoost can be expressed as:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (4)$$

where:

- $\hat{y}_i$: predicted value of sample i
- f_k : the k -th decision tree
- K : total number of decision trees

- x_i : input feature vector

Unlike linear regression, which constructs only a single prediction function, XGBoost builds multiple decision trees sequentially and combines their outputs to generate the final prediction.

The optimization process of XGBoost is based on the following objective function:

$$Obj = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (5)$$

where:

- $l(y_i, \hat{y}_i)$: loss function between actual and predicted values
- $\Omega(f_k)$: regularization term used to reduce model complexity

Regularization is one of the most important advantages of XGBoost because it helps reduce overfitting, which commonly occurs in tree-based models.

XGBoost was selected in this study for several important reasons.

First, XGBoost can effectively learn nonlinear relationships among input variables. In climate data, temperature does not follow a simple linear pattern but is influenced simultaneously by multiple environmental factors.

Second, XGBoost performs well on small and medium-sized datasets. The dataset used in this study contains approximately 1,400 daily observations, which is relatively suitable for machine learning algorithms.

Third, XGBoost reduces overfitting through several techniques:

- Regularization
- Shrinkage
- Tree pruning
- Subsampling

However, XGBoost is not specifically designed for time-series data because it cannot automatically learn sequential dependencies like recurrent neural networks.

Therefore, feature engineering was applied to transform time-series data into a tabular format suitable for the model.

The constructed features include:

Lag Features

$$Lag_1 = Y_{t-1} \quad (6)$$

$$Lag_7 = Y_{t-7} \quad (7)$$

$$Lag_{30} = Y_{t-30} \quad (8)$$

These features allow the model to capture the influence of previous observations on the current value.

Rolling Mean

$$RollingMean = \frac{1}{n} \sum_{i=1}^n Y_i \quad (9)$$

Rolling mean is used to represent short-term trends.

Rolling Standard Deviation

$$RollingStd = \sqrt{\frac{\sum (Y_i - \bar{Y})^2}{n}} \quad (10)$$

Rolling standard deviation is used to measure the level of variation in the data.

Additional time-based features were also included:

- Day
- Month
- Year
- Day of Week

These variables help the model learn seasonal and temporal patterns in the data.

XGBoost serves as an intermediate model between traditional statistical methods and Deep Learning approaches in this study.

3.4 Deep Learning Models

Although statistical and Machine Learning models can achieve good performance in many situations, they still have limitations in learning long-term dependencies within time-series data.

In climate data, the current temperature may depend not only on recent observations but also on meteorological events occurring several weeks earlier. For example, the effects of monsoon winds or atmospheric pressure changes may persist over a long period.

To address this limitation, Deep Learning models based on recurrent neural networks are employed to directly learn sequential structures in the data.

Before feeding data into the model, it is transformed into the following structure:

$$(samples, timesteps, features) \quad (11)$$

where:

- samples: number of data samples
- timesteps: number of previous time steps
- features: number of input variables

For example:

$$(1000, 30, 4) \quad (12)$$

This means that the model uses 1,000 samples, where each sample contains information from the previous 30 days and four meteorological variables.

3.4.1 SimpleRNN Model

SimpleRNN is the most basic form of recurrent neural network.

$$h_t = f(W_h h_{t-1} + W_x x_t + b) \quad (13)$$

where:

- h_t : hidden state at time t
- x_t : input data
- W_h, W_x : weight matrices
- b : bias

SimpleRNN uses information from previous hidden states to generate the current state.

However, when sequences become longer, SimpleRNN often suffers from the vanishing gradient problem, reducing its ability to learn long-term dependencies.

3.4.2 LSTM Model

LSTM was developed to solve the vanishing gradient problem of SimpleRNN.

Unlike SimpleRNN, LSTM uses memory cells and control gates.

The model consists of:

- Forget Gate
- Input Gate
- Output Gate

Forget gate:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (14)$$

Input gate:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (15)$$

Memory state update:

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t \quad (16)$$

LSTM allows the model to decide which information should be retained and which information should be discarded.

This capability enables the model to preserve long-term dependencies within the data.

3.4.3 GRU Model

GRU was developed as a simplified version of LSTM.

Instead of using three gates, GRU only uses:

- Update Gate
- Reset Gate

Update gate:

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \quad (17)$$

Reset gate:

$$r_t = \sigma(W_r[h_{t-1}, x_t]) \quad (18)$$

New hidden state:

$$h_t = (1 - z_t)h_{t-1} + z_t\tilde{h}_t \quad (19)$$

Because GRU contains fewer parameters, it generally trains faster than LSTM and reduces the risk of overfitting. Given the dataset of approximately 1,400 observations used in this study, GRU is expected to be more suitable than highly complex deep learning models.

3.5 Comparative Analysis of Forecasting Models

Experimental results indicate that there are considerable differences in performance among the forecasting models.

Statistical models such as SARIMA and VAR perform well when the data exhibits clear trends and seasonality patterns. The prediction curves generated by these models generally follow the actual observations closely during stable periods.

XGBoost demonstrates good adaptability to local fluctuations because of its ability to capture nonlinear relationships among variables. Through feature engineering techniques and tree-based learning mechanisms, the model can effectively identify hidden patterns in the data.

Meanwhile, Deep Learning models such as LSTM and GRU show advantages in learning long-term dependencies within sequential data. These models are capable of capturing complex temporal patterns that may not be easily modeled using traditional approaches. However, they typically require larger datasets to fully exploit their learning capabilities and achieve optimal performance.

One important finding of this study is that a more complex model does not necessarily produce better forecasting results. Prediction performance depends significantly on factors such as dataset size, data structure, and the assumptions underlying each model.

3.6 Chapter Summary

This chapter has presented a detailed description of the forecasting models applied in this study, including traditional statistical methods, Machine Learning techniques, and Deep Learning approaches. Each model was analyzed based on its operating principles, advantages, limitations, and applicability to climate data.

The use of multiple forecasting approaches creates a comprehensive evaluation environment, allowing the identification of the most suitable method for temperature forecasting in Delhi. The models developed in this chapter provide the foundation for experimental evaluation and discussion presented in the following chapter.

4 RESULTS AND DISCUSSION

4.1 Introduction

After completing the processes of data collection, preprocessing, stationarity testing, feature engineering, and forecasting model training, the next stage of this research was to evaluate the experimental results in order to determine the effectiveness of each method in the temperature forecasting problem. This stage plays an important role because it not only reflects the predictive capability of the models but also provides a basis for understanding the learning behavior of different algorithms when applied to real climate data.

The objective of this chapter is not simply to present the error values obtained from the models but also to analyze the underlying reasons for performance differences among forecasting methods. In many situations, a model with higher prediction accuracy is not necessarily the most appropriate model if its mechanism cannot be interpreted or if it lacks generalization capability when applied to real-world data. Therefore, evaluation should be conducted from both quantitative and qualitative perspectives.

Climate forecasting is considered one of the most challenging problems in time-series analysis due to the specific characteristics of meteorological data. First, temperature data often exhibit long-term trends and recurring seasonal patterns. In addition, meteorological variables such as humidity, wind speed, and atmospheric pressure do not operate independently but instead involve complex interactions. Furthermore, climate data often contain noise, irregular fluctuations, and various hidden exogenous factors.

Due to these characteristics, a model that performs effectively on conventional forecasting tasks may not necessarily achieve similar performance on climate datasets. Therefore, this study employs multiple evaluation criteria to provide a comprehensive assessment of each forecasting model.

In this study, forecasting performance was evaluated using several metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2).

MAE was used to measure the average absolute difference between predicted values and actual values. This metric provides a direct and easily interpretable representation of prediction error.

RMSE was employed to emphasize larger prediction errors by squaring deviations before averaging them. Therefore, RMSE is particularly useful in climate forecasting because large prediction errors in temperature forecasting may have significant impacts on decision-support systems.

MAPE represents prediction error as a percentage, allowing relative comparison across different models.

Meanwhile, the coefficient of determination (R^2) reflects the proportion of data variance explained by the model. Values closer to 1 indicate better representation of the underlying data

structure.

Besides quantitative evaluation metrics, this research also performed visual analysis through prediction plots and residual analysis to gain a better understanding of model behavior during forecasting.

4.2 Performance Comparison Between Models

Table 4 presents the forecasting performance of all implemented models, including Statistical models (ARIMA, SARIMA, VAR), the Machine Learning model (XGBoost), and Deep Learning models (SimpleRNN, LSTM, GRU). Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2).

Table 4: Forecasting Performance Comparison of All Models

Model	MAE	RMSE	R^2
VAR	1.3238	1.6528	0.9284
SARIMA	1.3180	1.6564	0.9280
ARIMA	1.3460	1.6818	0.9258
XGBoost	1.3151	1.7223	0.9222
GRU	1.9468	2.3613	0.8538
LSTM	2.2174	2.6376	0.8176
SimpleRNN	2.2305	2.6456	0.8164

The experimental results reveal substantial differences in forecasting performance among the implemented approaches. Overall, traditional statistical models achieved the best forecasting accuracy, particularly VAR and SARIMA, both of which obtained R^2 values above 0.92. These findings indicate that the models successfully captured the majority of variability in the temperature series.

Among all models, the VAR model achieved the highest R^2 value (0.9284), while SARIMA produced the lowest RMSE among the statistical models. The strong performance of VAR suggests that incorporating multiple meteorological variables such as humidity, wind speed, and atmospheric pressure significantly improves forecasting capability. Unlike univariate approaches, VAR models interactions among variables simultaneously, allowing the model to exploit more climate-related information.

SARIMA also demonstrated highly competitive performance because it explicitly models seasonal structures within the data. Delhi climate data exhibit strong yearly seasonal cycles, with recurring temperature increases during summer and decreases during winter. The seasonal differencing mechanism in SARIMA enabled the model to effectively learn these repeating patterns.

ARIMA achieved slightly lower performance compared with SARIMA. This outcome is expected because ARIMA does not explicitly incorporate seasonal components. Although ARIMA successfully learned short-term temporal dependencies, it was less effective in capturing long-term recurring seasonal behavior.

For the Machine Learning category, XGBoost achieved relatively strong performance with an R^2 value of 0.9222. The model successfully learned nonlinear relationships through feature engineering techniques such as lag variables and rolling statistics. Despite not being specifically designed for sequential forecasting, XGBoost demonstrated strong adaptability and competitive

forecasting accuracy.

In contrast, Deep Learning models produced lower forecasting performance compared with the statistical and Machine Learning approaches. Among the Deep Learning architectures, GRU performed better than LSTM and SimpleRNN, achieving an R^2 value of 0.8538. This result is consistent with theoretical expectations because GRU contains fewer parameters and generally performs better on medium-sized datasets.

Meanwhile, LSTM and SimpleRNN obtained the lowest forecasting accuracy. Although these models are theoretically capable of learning long-term dependencies and complex nonlinear relationships, the dataset used in this study contained only approximately 1,400 observations, which may not be sufficient for Deep Learning models to fully exploit their learning capacity.

The findings suggest that increasing model complexity does not necessarily guarantee better forecasting performance. For datasets with relatively clear seasonality and moderate sample sizes, traditional statistical models may remain more effective, computationally efficient, and interpretable than highly complex Deep Learning architectures.

4.3 Visual Analysis of Forecasting Results

Figure 8 illustrates the comparison between actual temperature values and the predictions generated by all forecasting models.

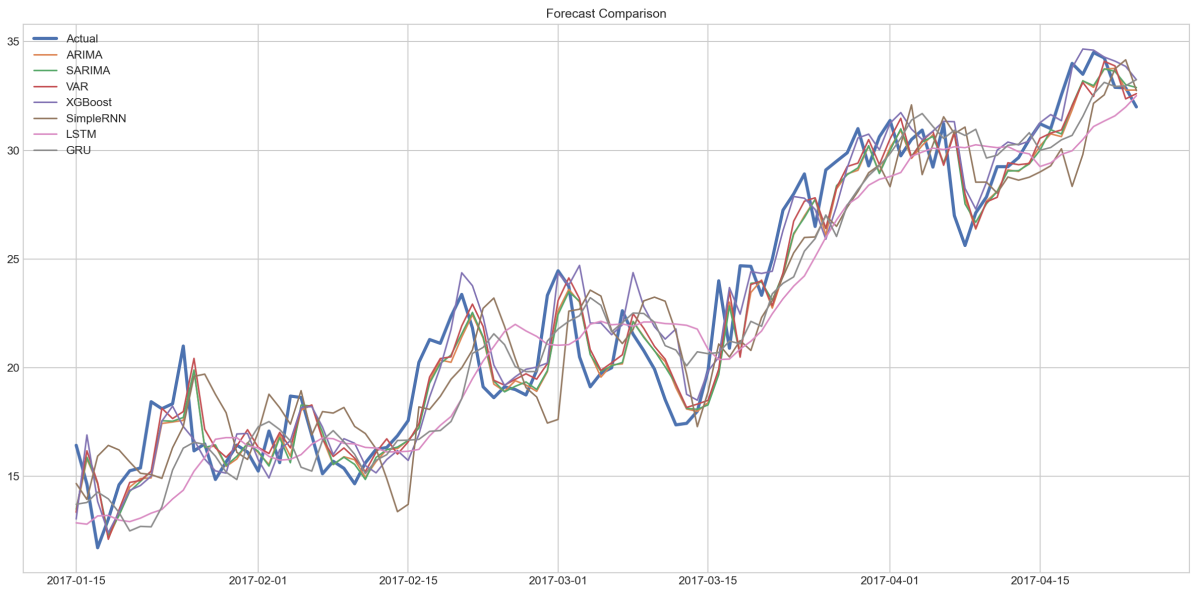


Figure 8: Forecast comparison between actual temperature values and predictions generated by all forecasting models. Statistical models such as VAR and SARIMA closely follow the actual observations, while Deep Learning models exhibit larger deviations during abnormal fluctuations.

The visualization results further confirm the quantitative findings presented in Table 4. Most forecasting models successfully captured the overall trend and seasonal structure of the temperature series. However, important differences emerged when examining local fluctuations and abrupt temperature changes.

The statistical models, particularly VAR and SARIMA, generated prediction curves that closely followed the actual observations throughout the forecasting horizon. These models effectively captured both the long-term seasonal cycle and short-term temperature fluctuations.

The prediction curves appeared relatively stable and smooth, indicating strong generalization capability.

The VAR model showed excellent adaptability during periods where temperature changes were influenced by interactions among humidity, wind speed, and atmospheric pressure. By modeling multivariate dependencies, VAR produced forecasts that aligned closely with actual climate behavior.

Similarly, SARIMA demonstrated strong capability in capturing repeating seasonal patterns. During seasonal transition periods, such as the shift between winter and summer, SARIMA forecasts remained highly consistent with observed temperature variations.

XGBoost also demonstrated competitive visual forecasting performance. Compared with the statistical models, XGBoost responded more flexibly to local fluctuations because of its ability to learn nonlinear relationships. The model effectively captured short-term variations and irregular movements within the data.

In contrast, the Deep Learning models exhibited several forecasting limitations. Although GRU, LSTM, and SimpleRNN successfully learned the overall trend of the series, their prediction curves frequently showed delayed responses during abrupt temperature changes.

LSTM and SimpleRNN particularly struggled during periods of sudden fluctuations, where predicted values deviated noticeably from actual observations. This delayed forecasting behavior suggests that the models were unable to fully learn the underlying temporal structure from the available training data.

The visual analysis therefore supports the conclusion that statistical models remain highly effective for climate datasets characterized by strong seasonality and moderate sample sizes. The results also indicate that more sophisticated architectures do not necessarily outperform simpler approaches when the dataset size is limited.

4.4 Residual Analysis

Residual analysis was conducted to further evaluate forecasting quality and identify potential weaknesses in model behavior. Residuals were calculated as the difference between actual values and predicted values:

$$Residual = Actual - Predicted \quad (20)$$

In an effective forecasting model, residuals should be randomly distributed around zero without exhibiting systematic patterns or strong temporal dependencies.

To avoid excessive redundancy, residual plots were selected from representative models of two groups:

- SARIMA and VAR representing the statistical model group
- LSTM and GRU representing the Deep Learning group

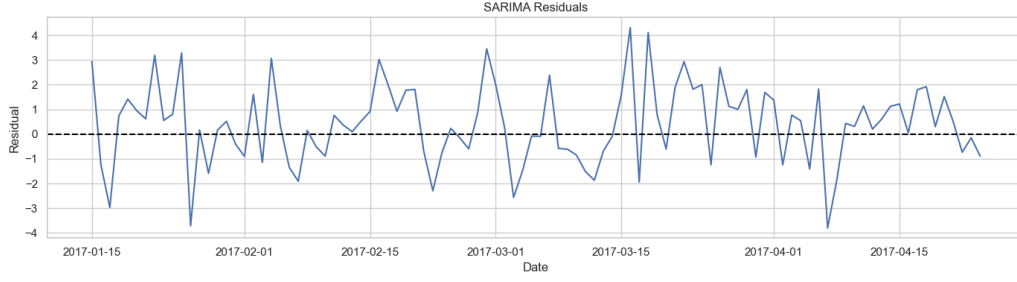


Figure 9: Residual distribution of the SARIMA model (unit: $^{\circ}\text{C}$). The residuals are randomly distributed around zero, indicating that the model successfully captured most of the important temporal and seasonal information contained in the dataset.

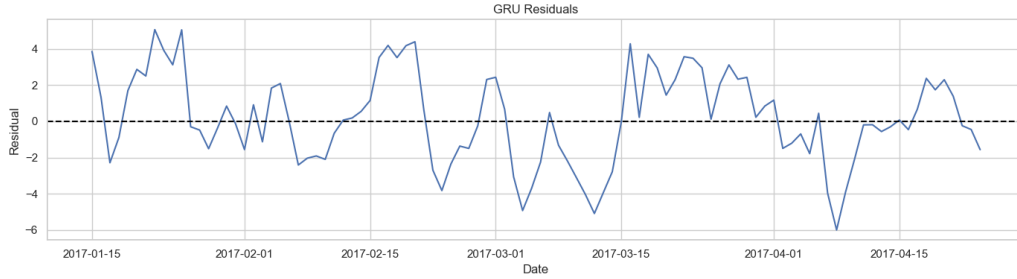


Figure 10: Residual distribution of the GRU model (unit: $^{\circ}\text{C}$). Although the residuals are generally centered around zero, several concentrated error regions and larger deviations are still observed, suggesting limitations in learning complex temporal structures from the available dataset.

The residual analysis results indicate clear differences between the forecasting approaches.

For the statistical models, residuals were generally distributed randomly around zero with relatively stable variance. This behavior suggests that the models successfully captured most of the meaningful information contained in the temperature series. No strong residual trends or repetitive structures were observed, indicating that the remaining forecasting errors behaved similarly to random noise.

VAR and SARIMA also produced fewer extreme residual values compared with Deep Learning models. This finding further confirms the strong forecasting capability of statistical approaches for structured climate data with stable seasonal behavior.

In contrast, the residual distributions of the Deep Learning models exhibited larger variability and several concentrated error regions. GRU performed more stably than LSTM and SimpleRNN, but residual clustering was still visible at certain periods.

These observations suggest that Deep Learning models were unable to fully eliminate hidden temporal dependencies within the dataset. One possible explanation is the relatively limited dataset size, which restricted the ability of neural networks to learn complex nonlinear relationships effectively.

Additionally, several large residual values appeared during abnormal weather fluctuations. Such forecasting errors may be associated with sudden climate events including heatwaves, rainfall changes, or atmospheric instability that cannot be fully inferred from historical observations alone.

Overall, the residual analysis confirms that the statistical models produced more reliable and

stable forecasting behavior under the characteristics of the current climate dataset.

4.5 Discussion

The findings of this study provide several important insights regarding climate forecasting using Statistical, Machine Learning, and Deep Learning approaches.

First, the experimental results demonstrate that model complexity does not necessarily guarantee superior forecasting performance. Despite the growing popularity of Deep Learning methods, traditional statistical approaches such as VAR and SARIMA achieved the best forecasting accuracy in this study.

One important reason is related to the characteristics of the dataset. The climate data used in this research exhibited relatively strong seasonality, stable temporal structures, and moderate dataset size. Under these conditions, statistical models are particularly effective because they are specifically designed to model trends, seasonality, and temporal dependencies.

In contrast, Deep Learning models such as LSTM and GRU generally require substantially larger datasets to fully exploit their capability of learning nonlinear and long-term dependencies. With only approximately 1,400 observations available for training, the neural network architectures may not have received sufficient information to learn complex climate dynamics effectively.

Another important finding is the strong performance of the VAR model. Unlike univariate approaches, VAR jointly models multiple meteorological variables, including humidity, wind speed, and atmospheric pressure. This multivariate structure enables the model to capture interactions among environmental factors, thereby improving forecasting quality.

The results also demonstrate the practical importance of proper model selection. In many real-world forecasting problems, simpler models may provide better trade-offs between forecasting accuracy, computational efficiency, and interpretability.

From a practical perspective, the findings suggest that statistical forecasting systems remain highly valuable for climate-related applications, particularly when datasets are moderate in size and contain strong seasonal patterns. Compared with Deep Learning architectures, models such as SARIMA and VAR require lower computational resources while still delivering strong predictive performance.

Furthermore, the study highlights the importance of understanding data characteristics before selecting forecasting approaches. Instead of prioritizing highly complex AI architectures, researchers should carefully evaluate whether the dataset structure aligns with the assumptions and requirements of the selected model.

Finally, the research provides empirical evidence that forecasting performance depends not only on algorithm sophistication but also on factors such as:

- Dataset size
- Seasonal structure
- Feature engineering quality
- Presence of exogenous variables
- Model suitability for the forecasting problem

These findings contribute important practical insights for future climate forecasting research and support the development of more efficient forecasting systems for environmental decision-making applications.

5 CONCLUSION AND FUTURE DIRECTIONS

5.1 Introduction

This chapter summarizes the major findings obtained throughout the research process on climate time series forecasting using Statistical, Machine Learning, and Deep Learning approaches. The conclusions presented in this chapter are derived from comprehensive experimental evaluations conducted using both quantitative performance metrics and qualitative visual analyses. Through the integration of forecasting accuracy evaluation, residual diagnostics, and time series structure analysis, this study provides a comprehensive understanding of how different forecasting approaches behave under the characteristics of real-world climate datasets.

Unlike many studies that focus solely on maximizing prediction accuracy, this research aimed to establish a complete and systematic forecasting framework covering all major stages of the forecasting pipeline. These stages include data collection, preprocessing, exploratory data analysis, stationarity investigation, feature engineering, model construction, training, validation, residual analysis, and comparative evaluation. Developing such a unified framework improves research reproducibility, methodological consistency, scalability, and adaptability for future forecasting applications.

Furthermore, this chapter discusses the broader implications of the research findings from both theoretical and practical perspectives. In addition to presenting the main contributions of the study, the chapter critically examines the limitations of the current research and proposes several promising future directions for improving climate forecasting systems.

5.2 Conclusion

Climate forecasting has become increasingly important in recent years due to the growing impact of climate change, urbanization, and the rising frequency of abnormal weather events worldwide. Accurate temperature forecasting systems are valuable not only for scientific research but also for practical applications involving agriculture, transportation, public health, energy management, environmental monitoring, and disaster prevention planning.

In this study, a comprehensive forecasting framework was successfully developed to predict temperature variations using historical climate data collected from Delhi, India, during the period from 2013 to 2017. The dataset included four important meteorological variables consisting of mean temperature, humidity, wind speed, and atmospheric pressure. These variables were selected because they represent major atmospheric conditions that directly or indirectly influence temperature dynamics over time.

Prior to model development, the dataset underwent extensive preprocessing and exploratory analysis procedures to ensure data quality and improve forecasting reliability. The preprocessing stage involved missing value verification, normalization, stationarity investigation, outlier inspection, correlation analysis, and time-series-oriented feature engineering. These procedures were essential because forecasting performance in time series problems depends heavily on data

quality, temporal consistency, and the ability of the models to capture underlying structural patterns.

The exploratory analysis revealed several important characteristics of the climate dataset. First, the temperature series exhibited strong seasonality with highly repetitive annual temperature cycles. Second, autocorrelation analysis demonstrated significant temporal dependencies between consecutive observations, indicating that historical temperature values strongly influence future observations. Third, correlation analysis showed that temperature variations were associated with interactions among multiple meteorological variables, particularly humidity and atmospheric pressure.

These findings confirmed that the forecasting problem cannot be treated as a purely random prediction task. Instead, temperature forecasting requires models capable of learning temporal structures, seasonal cycles, and inter-variable dependencies simultaneously.

To evaluate forecasting capability comprehensively, three major categories of forecasting approaches were implemented and compared under the same experimental conditions. The first category included traditional statistical models consisting of ARIMA, SARIMA, and VAR. The second category included the Machine Learning approach represented by XGBoost. The third category consisted of Deep Learning architectures including SimpleRNN, LSTM, and GRU.

The experimental results demonstrated clear performance differences among these forecasting approaches. Traditional statistical models, particularly VAR and SARIMA, achieved the best forecasting performance overall. Both models produced high R^2 values above 0.92 and generated prediction curves that closely followed actual temperature observations throughout the forecasting horizon.

The superior performance of SARIMA can largely be explained by the strong seasonal structure observed within the dataset. Since SARIMA explicitly incorporates seasonal autoregressive and differencing components, the model was highly effective in learning repeating annual temperature patterns. This capability enabled SARIMA to maintain stable and accurate forecasts across seasonal transitions.

Similarly, the strong performance of the VAR model highlights the importance of multivariate forecasting in climate prediction problems. Unlike univariate approaches that rely solely on historical temperature observations, VAR simultaneously models interactions among multiple meteorological variables. By capturing dependencies between temperature, humidity, wind speed, and atmospheric pressure, VAR successfully exploited additional climate-related information that improved forecasting accuracy.

The Machine Learning model XGBoost also demonstrated competitive forecasting performance. Through feature engineering techniques such as lag variables and rolling statistics, XGBoost was able to learn nonlinear relationships and short-term fluctuations effectively. Although its performance was slightly lower than the best statistical models, XGBoost showed strong flexibility and adaptability for handling structured climate data.

In contrast, Deep Learning models including SimpleRNN, LSTM, and GRU produced lower forecasting performance compared with the statistical approaches. Among the neural network architectures, GRU achieved the most stable performance, outperforming both LSTM and SimpleRNN. This outcome is consistent with theoretical expectations because GRU contains fewer trainable parameters and generally performs more efficiently on datasets with moderate sample sizes.

One of the most important findings of this study is that increasing model complexity does not

necessarily guarantee superior forecasting performance. Despite the popularity of Deep Learning models in recent years, the experimental results indicate that traditional statistical models may still outperform more sophisticated neural network architectures under certain data conditions.

Several factors help explain this observation. First, the dataset used in this study contained only approximately 1,400 daily observations, which is relatively limited for Deep Learning applications. Neural networks typically require substantially larger datasets to effectively learn complex nonlinear relationships and avoid overfitting problems.

Second, the climate dataset exhibited relatively stable and predictable seasonal behavior. Such characteristics align particularly well with the assumptions underlying traditional statistical models, especially SARIMA. When seasonality is strong and highly repetitive, statistical forecasting methods often achieve excellent performance with significantly lower computational complexity.

The residual analysis conducted in the previous chapter further confirmed these findings. Statistical models produced residual distributions that fluctuated randomly around zero with relatively stable variance, suggesting that the models successfully extracted most meaningful information from the dataset. In contrast, Deep Learning models exhibited larger residual variability and several concentrated error regions, indicating that hidden temporal dependencies remained insufficiently captured.

From a practical perspective, the findings of this study provide important implications for real-world forecasting system development. The results suggest that model selection should not rely solely on algorithm sophistication or current technological trends. Instead, forecasting models should be selected based on the characteristics of the dataset, the amount of available data, interpretability requirements, computational resources, and the practical objectives of the forecasting task.

Overall, the study successfully achieved its primary objectives by developing and evaluating a complete climate forecasting framework while providing empirical evidence regarding the strengths and limitations of multiple forecasting approaches for time series climate data.

5.3 Research Contributions

This study provides several important contributions from methodological, experimental, and practical perspectives in the field of climate time series forecasting.

The first major contribution of this research is the development of a complete and systematic forecasting pipeline covering the entire forecasting workflow. Unlike studies that focus only on model implementation, this research integrated all major stages of the forecasting process into a unified framework, including data collection, exploratory data analysis, preprocessing, stationarity analysis, feature engineering, model training, evaluation, and residual diagnostics.

Establishing such an integrated pipeline improves reproducibility, consistency, and scalability for future forecasting research. The framework also reduces manual intervention during preprocessing and model evaluation, thereby minimizing potential human errors and improving experimental reliability.

Additionally, the modular structure of the framework enables future adaptation to other forecasting applications beyond temperature prediction. The same methodology can potentially be extended to rainfall forecasting, air quality monitoring, energy demand prediction, traffic forecasting, and other environmental time series problems.

The second major contribution of this study is the systematic comparison of multiple forecasting paradigms within a single experimental environment. Statistical models, Machine Learning approaches, and Deep Learning architectures were all evaluated using the same dataset, validation strategy, preprocessing procedures, and evaluation metrics. This unified experimental design ensured fair and objective performance comparisons across different forecasting approaches.

From an empirical perspective, the study also contributes important evidence regarding the relationship between dataset characteristics and forecasting model performance. The findings demonstrate that forecasting accuracy depends not only on model complexity but also on factors such as seasonality, temporal dependency structure, dataset size, and the compatibility between model assumptions and data behavior.

This contribution is particularly important because many recent forecasting studies tend to prioritize highly complex AI architectures without sufficiently analyzing whether the dataset characteristics actually support such complexity. The results of this research provide practical evidence that traditional statistical models may still achieve highly competitive or even superior performance when datasets exhibit stable seasonal structures and moderate sample sizes.

Another important contribution of this study lies in the extensive use of residual analysis and time-series-specific exploratory analysis. By incorporating ACF/PACF analysis, rolling statistics, lag plots, seasonality analysis, and residual diagnostics, the research provides deeper interpretability regarding model behavior and forecasting quality rather than relying solely on numerical evaluation metrics.

Finally, this study contributes a valuable practical reference for future climate forecasting research involving structured meteorological datasets. The findings provide useful insights for researchers, practitioners, and policymakers regarding the selection of forecasting models under different environmental and data conditions.

5.4 Research Limitations

Although the study achieved promising forecasting results, several limitations remain and should be considered when interpreting the findings. Identifying these limitations is important because it provides an objective assessment of the current research while establishing a foundation for future improvements.

One of the primary limitations of this study is related to dataset size. The dataset consisted of approximately 1,400 daily observations collected over a relatively short time period from 2013 to 2017. While this amount of data is generally sufficient for traditional statistical forecasting models, it remains relatively limited for Deep Learning architectures that typically require substantially larger datasets to learn complex nonlinear relationships effectively.

Because of the limited amount of training data, Deep Learning models may not have fully exploited their theoretical capability for learning long-term temporal dependencies and hidden nonlinear climate dynamics. This limitation likely contributed to the lower forecasting performance observed for LSTM and GRU models compared with SARIMA and VAR.

Another important limitation involves the scope of input variables included in the forecasting framework. The study utilized only four meteorological variables: mean temperature, humidity, wind speed, and atmospheric pressure. Although these variables are highly relevant to temperature forecasting, real-world climate systems are influenced by many additional environmental factors such as rainfall, solar radiation, cloud coverage, air pollution, geographic conditions, and

large-scale atmospheric phenomena.

The absence of these additional variables may reduce the ability of the forecasting models to capture certain abnormal or highly complex weather behaviors. Consequently, forecasting errors may increase during periods involving unusual climate conditions or abrupt environmental changes.

The study also faces limitations associated with abnormal climate events. Climate systems are inherently dynamic and often affected by rare extreme events such as heatwaves, storms, or sudden rainfall changes. Since such events occur relatively infrequently, the available dataset contained only a limited number of abnormal cases for model training.

As a result, the forecasting models may struggle to generalize effectively during highly unusual climate conditions. This limitation was evident in the residual analysis, where larger forecasting errors appeared during periods of sudden weather fluctuations.

Another limitation concerns model generalization across geographic regions. All experiments in this study were conducted using climate data collected exclusively from Delhi, India. Although Delhi exhibits diverse climate behavior and clear seasonal structures, its environmental characteristics may differ substantially from other regions with tropical, desert, coastal, or temperate climates.

Therefore, the forecasting performance observed in this study cannot necessarily be generalized to all climate conditions worldwide. Additional experiments involving datasets from multiple geographic regions would be necessary to fully evaluate the robustness and transferability of the proposed forecasting framework.

5.5 Future Directions

Based on the limitations identified in this study, several promising future research directions can be explored to improve forecasting performance and system robustness.

One important direction involves expanding the amount of historical climate data used for model training. Future studies may incorporate longer observational periods spanning multiple decades or utilize higher-frequency climate observations such as hourly measurements. Increasing dataset size would provide richer temporal information and may significantly improve the learning capability of Deep Learning architectures.

Another promising direction is the integration of multiple environmental and external data sources. Future forecasting systems may combine traditional meteorological variables with satellite imagery, air quality measurements, rainfall records, remote sensing information, GIS data, and real-time sensor networks. Incorporating additional environmental information may help forecasting models better represent complex atmospheric behavior.

Developing hybrid forecasting architectures also represents a valuable research direction. Hybrid models combining statistical methods with Deep Learning techniques may simultaneously exploit the strengths of both approaches. For example, statistical models may effectively capture linear seasonal structures, while neural networks may learn hidden nonlinear relationships and irregular fluctuations.

Potential hybrid architectures include ARIMA–LSTM, SARIMA–GRU, and XGBoost-based recurrent forecasting systems. Such approaches may improve forecasting accuracy, robustness, and generalization capability under diverse climate conditions.

In addition, future studies may explore more advanced Deep Learning architectures specif-

ically designed for time series forecasting. Transformer-based models, Temporal Convolutional Networks (TCN), Attention-based recurrent architectures, and Bidirectional LSTM models have recently demonstrated strong performance across various sequential prediction tasks.

These architectures may provide stronger capability for learning long-range dependencies and complex multivariate interactions compared with conventional recurrent neural networks. Applying such models to climate forecasting problems may further improve prediction accuracy, especially when larger datasets become available.

Another important future direction involves developing real-time forecasting systems capable of continuously updating predictions using streaming environmental data. Integrating online learning mechanisms and adaptive forecasting strategies may improve the practical applicability of climate forecasting systems for operational decision-making environments.

5.6 Final Remarks

Climate forecasting remains one of the most challenging problems in environmental data science because atmospheric systems continuously evolve through highly complex interactions among numerous environmental variables.

The findings of this study demonstrate that forecasting performance depends not only on algorithm sophistication but also on many additional factors including data quality, preprocessing procedures, seasonality strength, temporal dependency structures, dataset size, and compatibility between model assumptions and data behavior.

An important conclusion derived from this research is that no universally optimal forecasting model exists for every climate forecasting scenario. Each forecasting approach possesses its own strengths and limitations depending on the structure and characteristics of the underlying dataset.

Traditional statistical models remain highly valuable when dealing with structured datasets exhibiting strong seasonality and moderate sample sizes, while Machine Learning and Deep Learning approaches may become increasingly effective when larger and more complex datasets are available.

The forecasting framework developed in this study provides a strong and extensible foundation for future climate forecasting research and practical forecasting system development.

With the continuous advancement of big data technologies, artificial intelligence, sensor networks, and high-performance computing systems, future climate forecasting models are expected to achieve increasingly higher accuracy, adaptability, and reliability. Such developments may provide substantial support for environmental monitoring, policy planning, disaster prevention, and sustainable decision-making in the context of global climate change.

References

- [1] S. Rao, *Daily Climate Time Series Data*, Kaggle Dataset. Available at: <https://www.kaggle.com/sumanthvrao/daily-climate-time-series-data>
- [2] BBC News, *India floods: More than 140 people die after heavy rain*, 2019. Available at: <https://www.bbc.com/news/av/world-asia-india-49311458>

- [3] S. Moreno-Carbonell, E. F. Sánchez-Úbeda, and A. Muñoz, “Daily outdoor air temperature time series analysis for long-term energy forecasting under climate change in Europe,” *Energies*, vol. 13, no. 7, p. 1569, 2020. Available at: <https://doi.org/10.3390/en13071569>
- [4] S. Hokoi, M. Matsumoto, and T. Ihara, “Statistical time series model of solar radiation and outdoor temperature — Model identification by Kalman filter,” *Energy and Buildings*, vol. 15, no. 3–4, pp. 373–383, 1990. ISSN: 0378-7788. Available at: [https://doi.org/10.1016/0378-7788\(90\)90011-7](https://doi.org/10.1016/0378-7788(90)90011-7)
- [5] Wikipedia Contributors, *Climate of Delhi*. Available at: https://en.wikipedia.org/wiki/Climate_of_Delhi